

BHARAT ACADEMIX CODEQUEST 2026

VaaniSetu

AI-Powered Multilingual Legal Aid Interpreter
for Rural India

Innovate. Build. Impact.

Social Impact & AI

Maddiboina Triveni — Individual



THE REALITY NOBODY SEES



"Growing up in Andhra Pradesh, I have seen how language becomes a wall between people and their rights. In my community, official documents in English or formal Hindi are treated as something only educated people can understand — everyone else just signs and hopes."

— Maddiboina Triveni, B.Tech CSE, Andhra Pradesh

Why This Problem Matters

73%

of India lives in rural areas

far from legal aid

40Cr+

Indians interact with legal system annually

with zero language support

1:50K

ratio of legal aid lawyers to rural citizens

grossly inadequate

PROBLEM STATEMENT



Untranslatable Legal Documents

FIRs, court summons, bail orders, and property deeds are written in formal English or Hindi. Rural citizens — farmers, laborers, tribal communities — cannot read them.



Wrongful Acceptance of Charges

Citizens unknowingly sign false confessions, miss hearings, or forfeit land rights. They accept punishments they never deserved — because they didn't understand the document.



Massive Scale, Zero Solution

800+ dialects exist in India. Legal documents exist in exactly 1 language accessible to less than 15% of rural citizens. No app has ever solved this for legal contexts specifically.

PROPOSED SOLUTION

VaaniSetu — "Voice Bridge"

An AI-powered multilingual legal assistant that translates court documents, FIRs, bail orders, and legal notices into the user's native dialect — delivered in plain language via voice and text — so every Indian understands their legal rights.



Speak or Photograph

Citizen photographs or speaks their document. OCR + ASR extracts the legal text instantly.



22 Languages, 100+ Dialects

Translated into native dialect — Telugu, Bhojpuri, Gondi, Tulu — not just Hindi or English.



Plain Language Explanation

Not just translation — legal context interpreted. "This means police can arrest you without a warrant."



Offline + SMS Fallback

Works on ₹5,000 phones. Offline mode for villages. SMS for 2G or feature phone users.

HOW IT WORKS

1



Capture

User photographs legal document or speaks into phone in their dialect

2



Extract

OCR (Google Vision) or Whisper ASR extracts raw legal text

3



Translate

IndicTrans2 + fine-tuned LLM converts to plain-language native dialect

4



Explain

Voice + text output: what it means, what to do, legal rights

5



Connect

Links to nearest legal aid center, shares summary PDF

TECHNOLOGY STACK

FRONTEND

- React Native (iOS + Android)
- Voice-first UI for low-literacy users
- Offline-capable PWA fallback

AI / NLP CORE

- IndicTrans2 by AI4Bharat (22 Indian langs)
- OpenAI Whisper ASR (voice input)
- Fine-tuned LLM on Indian legal corpus

BACKEND

- Python + FastAPI
- Google Cloud Vision OCR
- Firebase Auth + Realtime Sync

OFFLINE + ACCESS

- TensorFlow Lite (on-device inference)
- Twilio SMS API (2G / feature phones)
- ReportLab PDF generation

EXPECTED IMPACT

40Cr+

Potential Users

22

Indian Languages

800+

Dialects Covered

₹0

Cost to User



Reduces Wrongful Defaults

Citizens who miss hearings due to confusion contribute to 4.5 crore pending cases. VaaniSetu reduces this directly.



Integrates with eCourts

India's eCourts API is public. VaaniSetu can plug into the existing national court infrastructure — no new bureaucracy needed.

PROJECT ROADMAP

Week 1–2

Core AI Engine

- Integrate IndicTrans2 + LLM pipeline
- Train on 500+ public-domain court docs
- Build OCR-to-translation pipeline

Week 3

Voice & UI

- Whisper-based voice input module
- React Native app with dialect selector
- Plain-language voice + text output

Week 4

Offline & SMS

- TFLite model compression
- Twilio SMS API for 2G users
- End-to-end testing pipeline

Week 5

Demo & Docs

- Test 5 real document types
- Demo video production
- Technical documentation

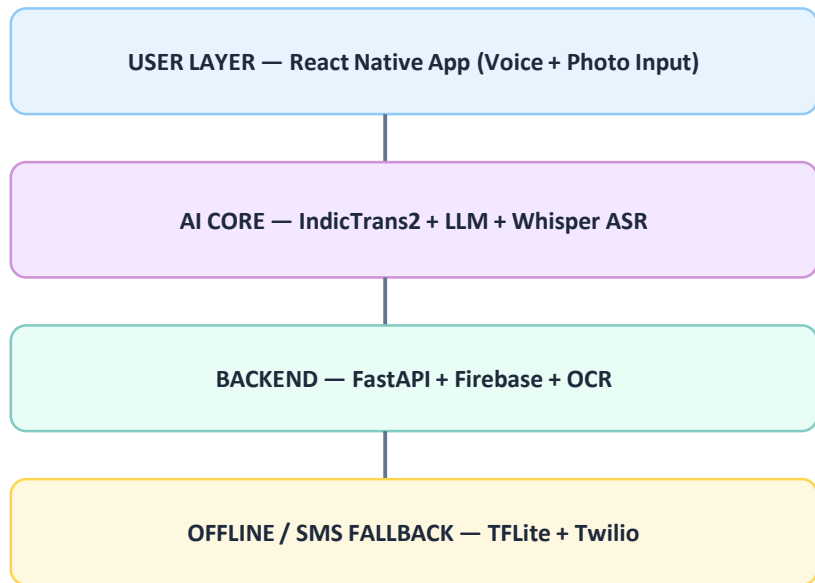
Post-Hack

Scale

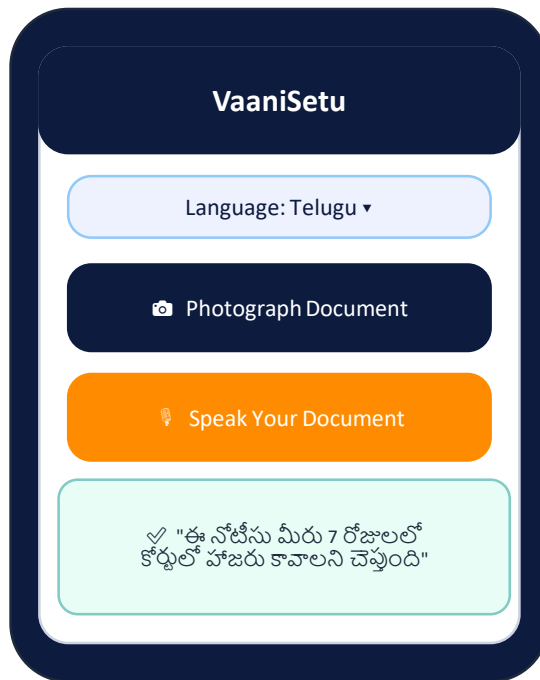
- DLSA partnership outreach
- eCourts API integration
- Open-source legal corpus

SYSTEM ARCHITECTURE & MOCKUP

Architecture



App Mockup



TEAM DETAILS

Maddiboina Triveni

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Name:	Maddiboina Triveni
Team Name:	trivenimaddiboina
Degree:	B.Tech — Computer Science & Engineering
Year:	Final Year (4th Year)
Participation:	Individual
Theme:	Social Impact & AI
Domain:	Legal Tech / Rural India

*"Justice delayed is justice denied. Justice untranslated is justice destroyed."
VaaniSetu bridges the gap — one voice, one village, one life at a time.*

